

2 Why can equal results lead to different scientific futures: How humans
3 and AI jointly form the conditions for further discovery

4 AI participates in mathematical research and sustained scientific work. How does historically
5 situated human-agent practice form capacities that change subsequent problem-solving
6 conditions? Mathematicians warn that AI's problem-solving gains can diverge from mathematical
7 goals: understanding, method development and transmission, training researchers and asking new
8 questions. OpenAI's Navier-Stokes account describes intermediate results organised into
9 conditions for later work; persistent agents make retaining, accessing, revising and using these
10 conditions operational.¹⁻³ Our account examines how relations between problems, methods,
11 evidence and responsibilities are formed and rebuilt; mathematics establishes insufficiency and
12 sufficient repair, while process records trace formation, revision and reuse. A finite construction
13 shows that equal specified current readouts need not determine the next correction: readout-only
14 correction has a positive lower bound on error, removed by a compact state-aware rule. Recorded
15 programmes undergo feedback-driven revision and changed-target reuse. Counterintuitively, one
16 revision relinquishes local exactness so a non-zero residual cancels an inherited-state term and
17 improves the whole: earlier work can create conditions requiring reconstruction of a previously
18 successful operation. Separately, in a selected known-proof completion, shared declarations become
19 a premise used downstream: with explicit researcher-written applicability guidance, the third
20 candidate is accepted in saved compilation; the comparison remains incomplete. Convergence,
21 unchanged versions and execution-dependent contrasts remain; no general history effect is
22 established. The separation and records distinguish obtaining results from sustaining their future
23 usability, making the formation, adoption, maintenance and reconstruction of research conditions
24 an explicit research object with identifiable people, records and tools, inspectable processes and
25 testable consequences.

26 Introduction

27 Mathematicians' statement warns that growing AI success in producing answers need not also form the
28 understanding, methods and capacities for further inquiry that mathematical practice seeks.¹ It treats
29 problem solving as a means towards understanding, training as a way of building understanding and the
30 ability to ask new questions, and explanation, refinement and transmission within the community as
31 conditions for methods and ideas to become usable by others.

32 OpenAI's published Navier–Stokes (NS) account illustrates a complementary relation: intermediate
33 results can become conditions for subsequent research through judgement, organisation and actual
34 adoption.² After the Euler result, the researchers selected NS as the most promising direction, redirected
35 agents from other problems and supplied that result to the continuing groups. Codex subsequently
36 consolidated insights across groups into guidance for further work. The account also reports an update to
37 a more extensively trained model during the effort, so these events describe an organised process with
38 several changing factors. In persistent Codex workflows, working files, notes, context transitions and tool
39 feedback make retaining and revising earlier work concrete operations.³ These settings expose a common
40 question: when people and AI produce useful results, what conditions do they also form for inquiry to
41 continue?

42 Our central question is **how humans and agents, through historically situated joint practice, form**
43 **capacities that change their own conditions of problem solving.** Our candidate answer concerns the
44 relations connecting problems, methods, evidence and responsibilities: research can form the conditions
45 for subsequent research, and these conditions can themselves require maintenance or reconstruction. The
46 same declared result need not determine how it can be checked, corrected or developed next. We establish
47 one precise limitation on a result-centred description: equality of a specified current readout can leave an
48 unavoidable discrepancy between the corrections required by different successor states. Preserving a
49 compact, relevant state removes that discrepancy. “Equal” names a specified product, observation or
50 current function, not a complete causal state; “scientific futures” names subsequent duties within stated

51 tools, permissions and resources. Compact preservation and legitimate reconstruction are positive
52 sufficient cases.

53 This distinction has established foundations. Proofs convey methodological knowledge beyond
54 statements, and productivity need not coincide with understanding.^{4,5} Retrieval-assisted proving studies
55 the availability of premises; truth-maintenance systems maintain reasons and revise dependent beliefs.^{6,7}
56 Predictive representations characterise state through future action-dependent consequences, and
57 equivalent expressive resources can have different optimisation dynamics.^{8,9} Organisational routines link
58 situated performances with patterns they preserve and change.¹⁰ These accounts already address use,
59 change and relationships. Our explanatory contribution is to organise the production, adoption and
60 reconstruction of research conditions as a connected object of comparison: a usable method interface, an
61 evidential obligation or an allocation of checking can be an outcome of earlier work and a condition of
62 later work. We operationalise this object by separating what is available from what is actually used, what
63 remains sufficient from what must be reconstructed, and a specified output from the state required for
64 continuation. The finite witness and the two kinds of process record make different parts of these
65 distinctions inspectable. Accordingly, an output comparison and a comparison of what can be done next
66 need not hold the same things fixed.

67 Recent studies make this distinction timely. Research on model warmth tests a relation between desirable
68 conversational properties rather than assuming their compatibility.¹¹ Model-collapse research examines
69 how generated outputs alter later training conditions.¹² Paper2Agent converts papers and computational
70 resources into interactive, executable research objects.¹³ Our concern is complementary: once a result or
71 tool is available, how is its applicability established, used and reconstructed when its consequences
72 expose a new requirement? We do not claim a newly discovered law of memory or proof application. We
73 make this relation inspectable through a finite non-sufficiency result, a constructive repair, actual
74 mathematical revision and successor-use records, and a known-proof application whose generated
75 premise is consumed downstream. The mathematical result specifies a distinction between comparisons;

76 the coefficient records make revision and changed-target reuse inspectable; and the Lean record shows a
 77 generated premise actually being consumed.

78 **Practice makes future conditions**

79 Consider a bounded research system containing people, agents, records, tools and the relevant
 80 environment. Let x_t denote its causally relevant state and $P_t = M(x_t)$ a specified present product or
 81 observation. Its recorded history H_t is evidence about how it reached that state, not a substitute for the
 82 state itself. A research arrangement links a problem to an admissible method, a method to evidence, and
 83 evidence to a participant or procedure authorised to use, challenge or revise it. These links have concrete
 84 carriers: parameter bindings, proof interfaces, test provenance, access rules, methodological explanations
 85 and assignments of responsibility. They may be explicit, distributed across participants or only partly
 86 recorded.

87 Four comparisons must therefore be separated. Product equivalence fixes a declared output or
 88 observation. Record equivalence fixes selected stored material. Complete causal-state equivalence fixes
 89 the relevant system and environment, rather than just their documentation. Finally, operational sufficiency
 90 is relative to a future regime F : specified duties, tools, permissions, resources and a declared family of
 91 continuation policies. Two states are equivalent under F when their relevant future outcomes and costs
 92 have the same law under those matched policies. A proof, an error value or a list of lemmas can be equal
 93 while the other comparisons are not. If complete causal states and transition rules are equal, an unencoded
 94 historical force cannot explain different futures.

95 A research episode begins with an arrangement that makes some actions possible: a problem formulation,
 96 methods that participants know how to use, evidence with an assigned scope, and responsibilities for
 97 checking it. Joint work produces more than a saved answer when it changes this arrangement. A tested
 98 estimate may justify choosing a new route; a proof may expose a reusable parameter substitution; a failed
 99 derivation may identify a missing hypothesis. The relevant carrier can be a formula, an executable file, an

100 explanation, a test receipt or a person's acquired method. Formation is identified by an actual construction
101 or revision history, not inferred merely because a useful object exists.

102 Adoption is the next distinct operation. A result becomes a working condition when it constrains a real
103 decision or is instantiated as the premise of another inference. The resulting action produces
104 consequences: a test confirms a limited range, a compiler rejects a binding, or a new application reveals
105 that the evidence did not cover the intended case. Such feedback can leave the method unchanged, narrow
106 its scope, alter an interface, or change who must check the claim. The changed arrangement becomes a
107 reproduced research condition only when it participates in another action. Figure 1 distinguishes the
108 established separation from its recorded operational instances. This sequence explains why accumulating
109 correct outputs and maintaining future usability can coincide, but need not do so.

110 The human organiser belongs inside this sequence. Marx's third thesis on Feuerbach challenges accounts
111 that place the educator outside the process of change.¹⁴ Mao's account of practice centres renewed action
112 and revision under changed conditions.¹⁵ Here these ideas determine the boundary of the research system:
113 selection, diagnosis, methodological explanation, checking labour and their possible errors must be
114 recorded alongside agent behaviour. They do not prove a computational mechanism. In the retained study,
115 human interventions altered the task and supplied an applicability strategy. Treating that contribution as
116 free, infallible background would misattribute jointly organised performance to the model alone.

117 Three operations have different evidential requirements. **U1**, using an existing method, includes applying
118 a known theorem with the right substitutions. **U2**, forming a method through practice, requires a newly
119 established procedure or relation with an identifiable production history; constructing a particular proof
120 object is a narrower instance than discovering a general method. **U3**, reconstructing a method under
121 changed conditions, requires a documented change in what the method must handle, a corresponding
122 reconstruction and evidence of subsequent use. These operations need not occur in order or in every
123 episode. A stable classical method may remain sufficient. A never-formed condition is not a lost one; an

124 available but unused method is not forgotten; and restoration of a previously available arrangement
 125 differs from first formation or new reconstruction.

126 The account yields three conditional propositions. First, if a later duty depends on an applicability relation
 127 not determined by a specified current product, product equality alone cannot warrant continuation
 128 equality; the mathematical example below provides one exact case. Second, if a preserved or
 129 reconstructed interface supplies all required observations, updates and operations within budget, a more
 130 elaborate historical carrier is unnecessary for that regime. Third, a claim that joint practice produced a
 131 capacity requires a trace of formation, actual use and relevant consequences, not just a correlation
 132 between archive size and a final score. These propositions change what a comparison holds fixed: current
 133 output, available information, adopted working relation and complete state are different controls. An
 134 intervention that changes a purportedly relevant relation but leaves the predicted consequence unchanged,
 135 without repair through another route, would narrow its proposed role. Stable classical methods and
 136 adequate single-system implementations are substantive sufficient cases, not failed controls.

137 **Equal readouts, different corrections**

138 The mathematical witness has a precise explanatory role: under its stated observation restriction, a present
 139 result does not determine the correction required next, whereas a compact state-aware operation does. A
 140 specified present readout is insufficient for exact correction across two admissible successor states. The
 141 source NS work uses a fixed primary wave and a covariance operator; positive amplitudes and an integral
 142 matrix support later signed corrections while retaining a quadratic remainder.¹⁶ We examine its explicitly
 143 derived coefficient reduction. Let z be the actual amplitude relative to a fixed base, p the target squared
 144 amplitude, $r = p - z^2$ the residual readout and g an applied increment. Writing $u = z - 1$, the complete
 145 consequence is

$$146 \quad E = (z + g)^2 - p = (2g + g^2 - r) + 2ug. \quad (1)$$

147 The first term is the pure correction residual at the fixed base; the second contains the effect of the state
 148 already produced. Reducing the first term need not reduce their sum. Thus a continuation cannot be
 149 described solely as solving the same fresh problem repeatedly: the previous operation changes what the
 150 next one must receive.

151 Fix $0 < r \leq 1/4$ and $0 < v \leq 1/8$, and consider $z_{\pm} = 1 \pm v$ with targets $p_{\pm} = z_{\pm}^2 + r$. These two states have the
 152 same declared readout r . Any deterministic action depending only on that readout must use the same g ,
 153 giving

$$154 \quad E_{\pm} = g^2 + 2(1 \pm v)g - r, E_+ - E_- = 4vg. \quad (2)$$

155 Both residuals cannot vanish: that would require $g = 0$ and hence $r = 0$. More strongly,

$$156 \quad \inf_{g \in \mathbb{R}} \max(|E_+|, |E_-|) = 2v(\sqrt{1+r} - 1) > 0. \quad (3)$$

157 To see the bound, write the maximum as $|g^2 + 2g - r| + 2v|g|$. On $g \geq 0$ its minimum occurs at
 158 $g = \sqrt{1+r} - 1$; on $[-2, 0]$ it is at least r , and reflection of the remaining negative branch adds a positive
 159 offset. Supplementary Note 2 gives the piecewise proof. The restriction is explicit: the same deterministic
 160 action sees only r , whereas $z_{\pm}$ and $p_{\pm}$ differ. This is not equality of complete current function, nor an
 161 assumption that the model in the later case ignored its state.

162 Adding the relevant state gives a constructive alternative. For $r \in [-1/2, 1/2]$ and $u \in [-1/4, 1/4]$, let
 163 $z = 1 + u$ and $w = \sqrt{z^2 + r}$. The radicand is at least $1/16$. Define smooth core functions

$$164 \quad C_*(r, u) = -\frac{1}{2z}, B_*(r, u) = -\frac{1}{2z(w+z)^2}. \quad (4)$$

165 With a smooth cutoff Φ equal to one on $|r| \leq 1/4$, $|u| \leq 1/8$ and zero outside the stated outer rectangle, the
 166 operation

$$167 \quad g = \frac{r}{2} + \Phi(r^2 B_* + ur C_*) \quad (5)$$

168 satisfies $g = w - z$ and $E = 0$ in the inner region. Outside the outer rectangle it returns to the signed linear
 169 correction; the transition remains smooth. The relevant state and a compact operation thus suffice for
 170 exact local correction. Positivity and all denominators are checked on an open neighbourhood of the
 171 stated outer rectangle, with no evaluation of a square root across a negative radicand.

172 This witness gives the title a precise meaning (Fig. 2): equality of a declared present quantity can coexist
 173 with a positive lower bound for observation-restricted correction, and the separation can disappear when
 174 the required state is preserved or reconstructed. These are standard algebraic tools applied to a specified
 175 continuation problem. Extending the construction to weighted wave classes requires the source derivative,
 176 support and multiplier assumptions. Neither the local witness nor a finite numerical check independently
 177 proves the global NS theorem, universal norm improvement or the properties of arbitrary generated code.
 178 Supplementary Note 2 states this mathematical scope once alongside the preserved counterexamples.

179 **When compact inheritance is enough**

180 A compact record can preserve continuation without preserving the entire history. Suppose a compact
 181 record $c = C(x)$ supports the observations, permitted actions, successor updates and costs required by a
 182 declared family of future duties. Under common joint kernels, matched exogenous-input laws and the
 183 initial-history conditions of Supplementary Note 3, factorisation through c and a common compact-state
 184 update reproduce the relevant finite continuation law. This is the standard sufficient-state composition
 185 principle, proved by induction in Supplementary Note 3 and consistent with predictive representations.⁸
 186 The result includes the transitions: reproducing today's answer alone is insufficient. Legitimate
 187 rederivation also supplies a sufficient route when its construction and retrieval costs fit the specified
 188 budget.

189 The scientific problem is consequently not to maximise retained history. It is to identify what must remain
 190 actionable as the duties change. A compact interface, a conventional method or a single integrated solver
 191 can be fully sufficient; copying an entire transcript may add no value. An archive can preserve
 192 information without making it usable at the relevant time. Conversely, a summary can preserve the right

conditions without preserving their entire production history. A shift to new duties can invalidate a previous sufficiency judgement, making the representation or method itself a legitimate object of further practice. This is the transition from inheritance to reconstruction, rather than an assumption of inevitable degradation.

Recorded revision and continued use

The coefficient records connect the distinction in equation (1) to actual formation, feedback-driven revision and subsequent use: an exact pure component can leave a whole residual through its interaction with the inherited state. Archived coefficient studies record actual revision and reuse under changed targets. Four formation episodes in an assistance comparison and eight in a version-adoption comparison produced 52 recorded stages and 104 completed model returns. Real programmes, feedback and the resulting amplitude fields were carried forward to changed targets. The common operation was $g = r/2 + \Phi(r, u)[r^2 B(r, u) + ur C(r, u)]$; the agent supplied B, C , while the investigators supplied the template, cutoff, sources and duties. These observations directly instantiate equation (1), but not a complete NS construction. Two post-selected traces make the mathematical content of this continuation explicit (Fig. 3; Supplementary Note 9).

In one assistance episode, the initially formed rule retained $B_0 = -1/[2(1 + \sqrt{1+r})^2]$ and $C_0 = 0$. Writing $s = \sqrt{1+r}$, it gives $g = s - 1$. At formation, the declared core was evaluated from the fixed base, so the pure residual also described the complete residual. In the subsequent duty, the actual state produced by that work was inherited rather than reset. The same rule still made the pure residual exactly zero, but its full successor residual was $2u(s - 1)$.

Within this first continuation, the initial, draft and final versions used the same incoming state and target at each matched coordinate. The agent's first revision introduced $C_1 = -1/[s(1+s)]$, changing the complete residual to $-u^2 r/(1+r)$. After numerical feedback, it produced a smooth secant expression for C , while retaining B_0 . On the declared core this gives $g = \sqrt{(1+u)^2 + r} - (1+u)$ and an exactly zero

217 complete residual. Crucially, the revised pure residual is generally non-zero: it cancels the inherited-state
 218 cross term. Correct continuation here required relinquishing a locally exact subresult, not improving every
 219 component separately.

220 The recorded next continuation received and retained that actual programme and used its predecessor's
 221 state for a changed target. The within-duty version comparison documents reconstruction; the subsequent
 222 duty documents inheritance and actual renewed use. Because the inherited state and target changed, the
 223 latter is not another comparison under the same conditions. The programme's availability and its adoption
 224 as a condition of further work are separately visible in this record.

225 A second episode recorded $B = -1/8$ and successive programmes with $C = -1/2, -1/2 + u/2$, and
 226 $-1/2 + u/2 - u^2/2$. These were generated, applied and passed onward, rather than inserted retrospectively
 227 by the analyst. For $u = a\tau$ and $r = b\tau^3$, their whole residuals generally begin at orders τ^5, τ^6 and τ^7 . The
 228 later duty was evaluated on the actual inherited field, not a reset base. This is a finite instance of revision
 229 under a changed target and renewed use. It is not a new general series method: the coefficients follow a
 230 familiar expansion, and the original explanation of one remaining term contained a sign error that is
 231 distinguished from the effective programme in Supplementary Note 9.

232 These traces are process evidence, not substitutes for the original comparisons. Seven of eight version
 233 pairs had identical initial and revised code; five final responses across the two studies were empty. In the
 234 fixed outer region, any valid programme applied $r/2$, whereas an empty final response applied no update.
 235 This route dominated large positive or negative aggregate contrasts. All original denominators and
 236 outcomes are retained, including the separately authorised supplementary observation replacing an
 237 unavailable analysis slot without resolving its historical unknown request. No stable average assistance or
 238 adoption benefit was established. Earlier finite-scalar comparisons also converged after adaptation. Thus
 239 revision and renewed use were observed, whereas their general causal advantage remains unresolved.
 240 This distinction replaces neither null results with selected successes nor executed work with a claim that
 241 no continuation occurred.

242 **An available result becomes a premise**

243 A separate proof-completion record shows an existing mathematical result being made usable by a
 244 supplied consumer, at a different location in the covariance construction. The recorded task completes
 245 `hclose` inside the supplied Lean theorem `compact_chart_pair_bounds`.^{17,18} The outer proof and two relevant
 246 lemma declarations were identical in the two conditions. An additional note in one condition explained
 247 how to apply those results to the current obligation. This deliberately selected, known-proof task supplies
 248 a local record of turning existing mathematics into an accepted premise; it is not a controlled instantiation
 249 of the two-state lower bound in equation (3).

250 The common mathematical assumption bounds a pulse-ratio error on an interval of length L_n . The
 251 available integral estimate expects the interval, denominator and centre in terms of R_n^2 . A second common
 252 lemma states $R_n^2 = L_n$. The obligation is therefore not to discover a missing numerical inequality, but to
 253 transport the entire dependent premise into the representation required by the integral estimate. Rewriting
 254 interval membership alone is insufficient: the conclusion obtained from that membership must also be
 255 rewritten. Figure 4 locates the exact application and its fixed downstream use.

256 After this transport, the supplied integral lemma yields

$$257 \quad |\overline{H}(P)_{ij} - H_0(\xi)_{ij}| \leq \frac{e + d c}{R_n} \leq \frac{e + d |c|}{R_n} = \frac{Q}{R_n} \leq \rho. \quad (6)$$

258 Here $d \geq 0$, $R_n > 0$, $Q = e + d |c| \geq 0$, and the last comparison is already provided. The symbol e denotes the
 259 source error coefficient, not the residual E in equation (1). No division by Q is required, so the case $Q = 0$
 260 is retained. The resulting `hclose` object is passed to the pre-written `hmargin` application, which supplies
 261 margins subsequently combined with column-scale bounds to construct `ZeroOrderBounds`. This is actual
 262 consumption of a local proof object by downstream code, not an inferred relevance relation. The pulse
 263 construction, integral estimate, compactness argument and downstream plan themselves were supplied.

Abstractly, a result $L : \forall \theta, A(\theta) \rightarrow C(\theta)$ becomes usable for a consumer requiring I only after an admissible parameter choice, evidence for $A(\theta)$ and any necessary transport from $C(\theta)$ to I have been supplied. This is ordinary proof composition, not a new logical mechanism. It specifies the concrete relation observed here: common declarations did not by themselves determine whether a usable instance was constructed within the available work sequence. A correct compact record of that instantiation could be sufficient. The empirical case concerns application, whereas equations (1)–(5) concern state-dependent correction. Their correspondence is explanatory, not a claim that the two experiments or causal mechanisms are identical.

What the recorded comparison establishes

The saved record contains six returned model requests and five compiled candidates (Fig. 4). The applicability-guided condition produced two rejected candidates, then a third accepted candidate. The declarations-only condition first reached the 32,768-token output limit without a visible proof body; its next two candidates were rejected. The accepted body was mechanically inserted into the common enclosing theorem, whose saved Lean output reports the exact local type and dependencies `propext`, `Classical.choice` and `Quot.sound`. Acceptance here denotes that retained compilation record; the local target’s meaning and the status of subsequent verification are stated separately in Methods.¹⁹

Both final candidates invoked the relevant integral lemma and scale identity. The final declarations-only body converted the interval-membership hypothesis, but returned an inequality still expressed using L_n where the expected type used R_n^2 . The accepted candidate completed the representation change and the scalar comparison. The difference is therefore narrower and more informative than “one condition found the theorem and the other did not”. It identifies where an available result failed to become a consumer-ready premise. The guided condition’s earlier failures are retained alongside this success; the first truncated comparison request is not labelled a mathematically refuted proof.

The investigators selected this gap after several broader workflows had been modified or stopped. The guided note was researcher-written procedural information, not a blind summary of history; it added

289 initial input tokens. Realised computation also differed: guided requests used 25,757 input and 20,781
290 output tokens, compared with 12,124 and 41,009, respectively, in the comparison. Compiler feedback and
291 subsequent inputs were consequently different. The experiment documents this local application sequence
292 and its recorded contrast. It does not establish a general success rate, efficiency advantage or causal effect
293 isolated from extra guidance and context use. Position and presentation are plausible alternatives in light
294 of previous context-use research.²⁰ No history-retrieval or compression manipulation occurred in the final
295 protocol.

296 The proof-completion episode records use of existing methods, formation of a particular proof object and
297 feedback-based revision of that object. Its downstream consumer was supplied, so it does not by itself test
298 reconstruction for a new duty. The coefficient episodes do contain changed-target revision and renewed
299 use, under an investigator-provided interface. These complementary records support the existence of
300 particular application and reconstruction processes, not autonomous formation of every condition in the
301 theoretical cycle. Their distinct inputs, tasks and sampling prevent their treatment as independent
302 confirmations of one general effect.

303 These distinctions also prevent several different histories from receiving the same explanation. An empty
304 terminal return records no final programme, not proof that a method was never formed; a draft may
305 already exist. Identical code can still be adopted and used on an inherited state. A failed conversion of an
306 existing lemma concerns applicability at the consumer interface, not absence of the lemma. Changed-
307 target revision followed by renewed use is documented in the coefficient traces; first non-formation, an
308 available method never adopted, and restoration of a lost arrangement remain distinct theoretical
309 possibilities rather than additional observed categories in these records. Formation of the accepted proof
310 object does not measure a change in human understanding.

311 In the coefficient trace, the earlier rule remained available; what changed was its relation to the inherited
312 state and target. The record therefore locates reconstruction of applicability and subsequent adoption,
313 rather than disappearance of the method.

314 **Conditions for further discovery**

315 The account makes the formation, adoption and reconstruction of usable research conditions an explicit
316 object of explanation and evaluation. Its established mathematical separation is specific and constructive:
317 a declared current readout need not determine the correction required next, and relevant state can provide
318 a compact sufficient repair. Actual records show both a formed programme being revised and reused
319 under changed targets and an available theorem being made into a premise consumed by downstream
320 code. In the former, improving the complete result could require abandoning exactness of a component.
321 Together these contributions connect an exact distinction between comparisons to inspectable operations
322 in joint practice. They make result production and the maintenance of usable research conditions
323 distinguishable objects of inquiry, without requiring long histories to be intrinsically valuable or a new
324 force beyond ordinary state and transition descriptions.

325 For mathematical understanding and training, the finite separation changes the comparison that matters.
326 Equal assessed answers do not establish that learners can check assumptions, recognise another use or
327 formulate the next problem. Evaluation can therefore pair a current result with a later duty requiring a
328 stated relation—such as adapting a method after a hypothesis changes—and record both. The human
329 prediction is that differences appear where the necessary relation is not fixed by the assessed product and
330 cannot be reconstructed within budget. If a compact explanation or ordinary rederivation restores transfer,
331 it is a successful educational design, not a failure to demonstrate historical necessity. If matched product
332 performance also predicts all prespecified continuation duties across such interventions, the proposed
333 additional distinction is unnecessary in that setting. Testing these claims requires actual human activity;
334 Lean acceptance is evidence about a proof object, not a measure of understanding.^{1,4,5}

335 For research organisation, an intermediate result has value both as a finding and as a possible change to
336 the next research arrangement. It can redirect a route, shift the evidential responsibility between
337 participants, or justify reallocating resources. Recording only the final output obscures the labour that
338 makes these changes workable. A contribution record should therefore distinguish production,

applicability work, actual adoption and continuing maintenance; it should also record unsuccessful coordination and corrections to human judgement. The NS process account motivates this distinction without separating the effects of transfer, model changes, resources and selection.² Organisational research can test this relation where duties and access can be documented. A handoff intervention can track changes in checking or research routes, then test whether another carrier restores the same outcome and cost. Equivalent compact handoffs demonstrate preservation of the relevant conditions; no change without such restoration narrows the proposed relation's role.¹⁰

For persistent agents, a completed task can be evaluated by what remains possible afterwards: correction under new evidence, adjustment of scope and reconstruction of a method for a changed objective. Pinned Codex sources make notes, context transitions and historical access distinct operations; their availability does not establish that the right conditions are preserved.³ Executable-paper systems provide another useful carrier.¹³ The design question is what state or application interface must survive, who can revise it, and whether the revision is used. A compact certified interface, reliable classical routine or integrated solver may satisfy these requirements without a long transcript. Conversely, preserving every message may still fail to make the relevant relationship actionable. These predictions require specified future duties, budgets and observations of restoration; they do not license inferring alienation or loss of agency from one failed proof attempt.

Written proofs, executable checks, preserved programmes and complete comparison records provide a basis for independent reproduction and extension. The present work is responsible for its finite claims, their assumptions and these traceable instances. Mathematical training can investigate how current achievement relates to later checking and method transfer; organisational research can track how adoption changes responsibilities and research routes; persistent-agent research can test what must remain actionable for correction and adaptation. Each extension has to specify its duties and costs and retain cases in which compact records, classical methods or rederivation suffice. The common question is not simply how many results a system produces, but what its work leaves people and agents able to do next.

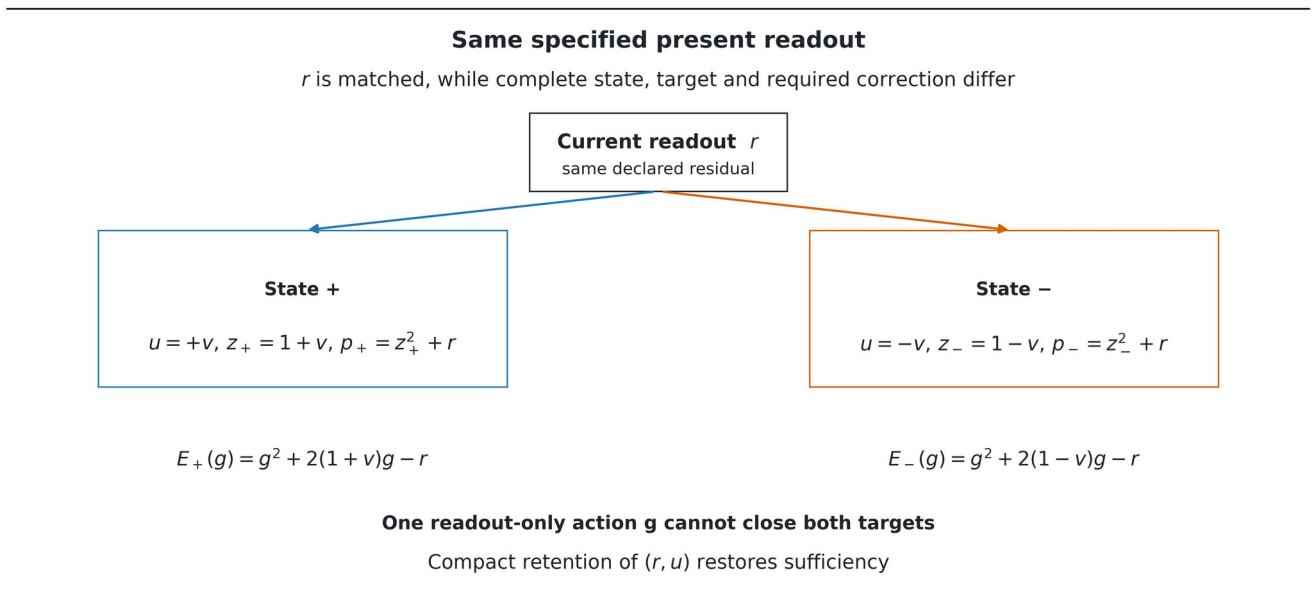
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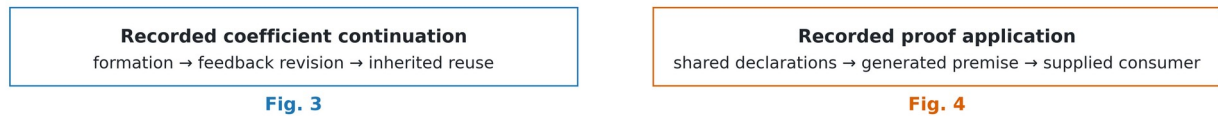
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411 Figure legends

a A specified present readout is not sufficient



b Two recorded entry points into common practice

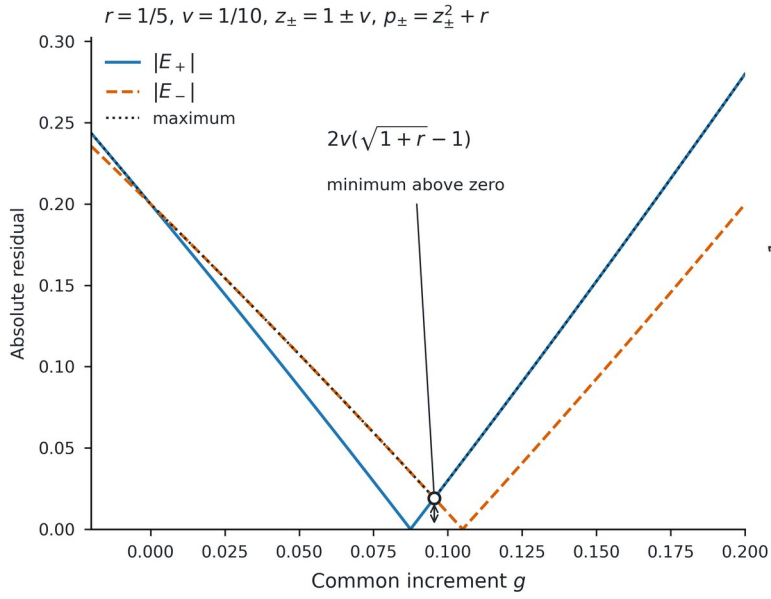
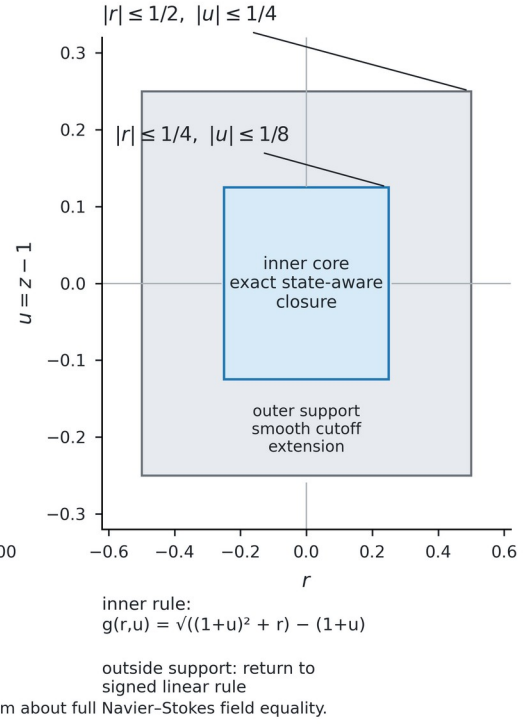


Humans, AI, feedback and adoption remain inside the research object

Evidence identities remain separate: theorem, successor records and Lean proof record are not one identified causal loop

412

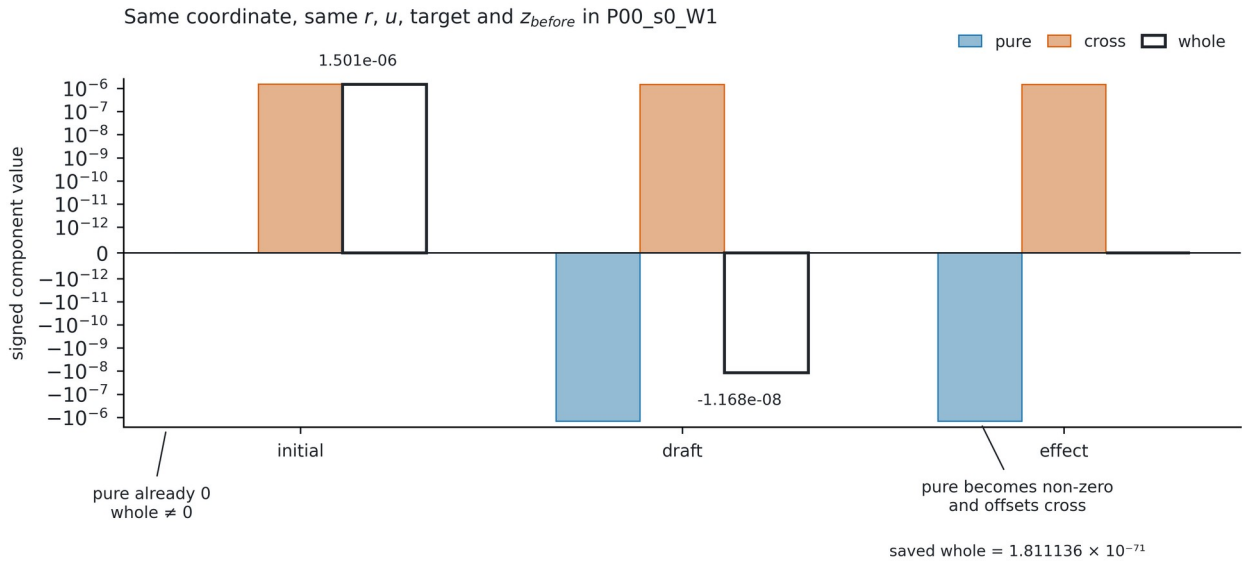
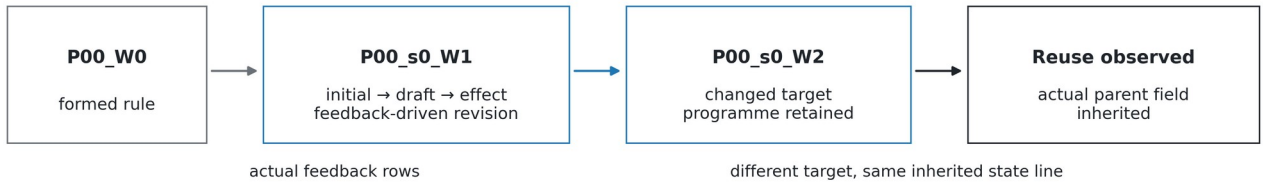
413 **Fig. 1 | A specified current readout is not a sufficient scientific state.** **a**, The analytical witness fixes the declared current
 414 readout r while allowing different complete states and targets. For $0 < r \leq 1/4$ and $0 < v \leq 1/8$, the two states have $u = \pm v$,
 415 $z_{\pm} = 1 \pm v$ and $p_{\pm} = z_{\pm}^2 + r$. A correction that depends only on r must be shared and cannot close both targets. Retaining the
 416 compact pair (r, u) repairs this non-sufficiency. The comparison concerns a specified present readout, not equality of
 417 complete causal states or an identified causal effect of history. **b**, Two archived entry points into common practice are then
 418 separated by evidence identity: coefficient continuation (Fig. 3) and proof application (Fig. 4). The theorem, the successor
 419 records and the Lean proof record are linked conceptually but are not merged into one identified causal loop.

a A positive minimax gap under a readout-only rule**b Compact repair and declared domain**

Analytical curves and declared domains only: no empirical sample size and no claim about full Navier-Stokes field equality.

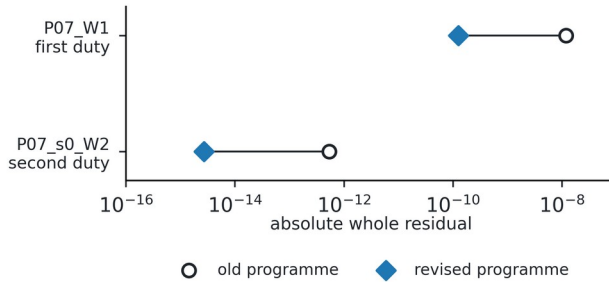
420

421 **Fig. 2 | Positive minimax separation and compact repair.** **a**, Curves show $|E_+|, |E_-|$ and their maximum for the exact
 422 witness at $r = 1/5$ and $v = 1/10$. The minimum of the maximum is the positive distance $2v(\sqrt{1+r}-1)$ above the zero
 423 line. These are analytical values, not experimental samples. **b**, The state-aware rule is exact on the inner rectangle $|r| \leq 1/4,$
 424 $|u| \leq 1/8$. A smooth extension is declared on the outer support $|r| \leq 1/2, |u| \leq 1/4$, and the signed linear rule operates
 425 outside the support. The panel displays domain structure rather than a statistical experiment.

a One revised programme improves the whole relation by giving up a locally exact component**b Actual feedback, parent-state inheritance and changed-target reuse**

$$B_0 = -1/[2(1 + \sqrt{1+r})^2], C_0 = 0$$

final C = smooth secant expression; B retained

c Further revision is compared within, not across, duties

52 recorded stages • 104 complete returns

7/8 adoption pairs kept identical code
• 5 empty final programmes

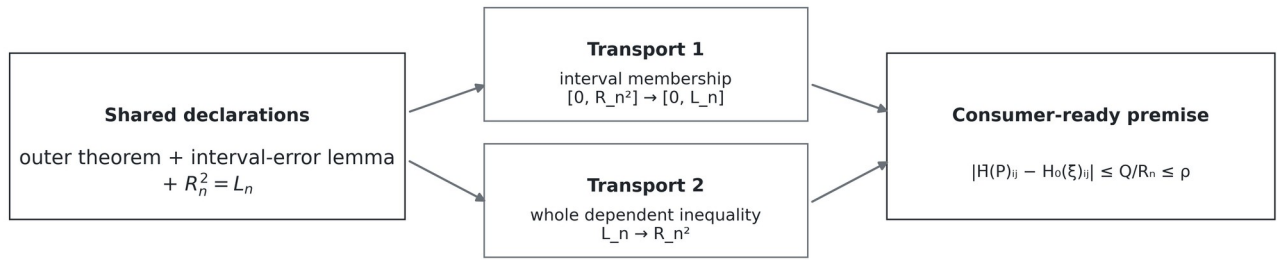
P07 points are post-selected explanatory traces;
they do not establish an average gain.

Do not join different targets into one learning curve.
The two duties use different inherited states and targets.

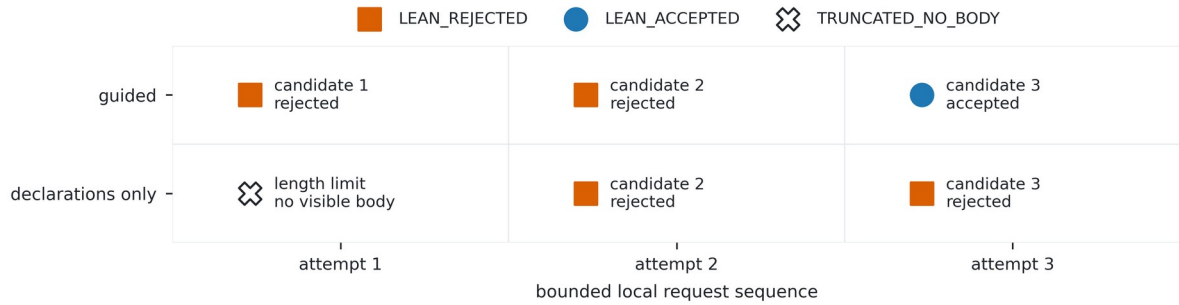
40 parent-state links were checked byte-for-byte.
Unchanged versions, failures and the supplementary
observation remain in Source Data.

426

427 **Fig. 3 | Actual programme revision, inheritance and renewed use.** **a**, In assistance episode P00_s0_W1, the same
428 coordinate, r , u , target and z_{before} are shown for the initial, draft and final effect versions. The displayed decomposition is
429 $pure = g^2 + 2g - r$, $cross = 2ug$, and $whole = pure + cross$. Initially, the pure term is exactly zero while the
430 whole residual is not. After revision, the pure term is no longer zero, but it cancels the cross term; the saved final whole
431 residual is retained as 1.811136×10^{-71} rather than replaced by an artificial zero. **b**, The process panel preserves actual
432 feedback-driven revision, programme identity, parent-state inheritance and changed-target reuse from P00_W0 through
433 P00_s0_W1 to P00_s0_W2. **c**, P07 compares old and revised programmes separately within two distinct duties. These are not
434 connected into one learning curve because the duties use different inherited states and targets. The figure retains the full-cohort
435 boundary: 52 recorded stages, 104 completed returns, 7/8 unchanged adoption versions, 5 empty terminal programmes, and the
436 boundary that these post-selected traces do not establish a general average benefit.

a A shared result must be transported into the consumer's representation

Rewriting interval membership alone is not enough; the entire dependent conclusion must also be transported

b The generated proof object is actually used downstream**c All six requests, five candidates and the truncated control request**

First declarations-only request was truncated and had no body; it was not a Lean rejection. Guidance added an application strategy and the downstream consumer was supplied.

This is one selected paired comparison: six requests and five candidates are not independent replications, and saved compilation acceptance is not upgraded here to new kernel certification.

437

438 **Fig. 4 | A generated proof object is made usable and actually used.** **a**, The proof first transports interval membership into
 439 the source hypothesis's representation, then transports the resulting dependent inequality back into the consumer's
 440 representation; membership transport alone is insufficient. **b**, The generated `hclose` is explicitly distinguished from the
 441 supplied downstream consumers `hmargin` and `ZeroOrderBounds`; the chain `generated hclose` → `supplied`
 442 `hmargin` → `ZeroOrderBounds` is an observed use relation. **c**, All six requests in the selected paired proof-completion
 443 comparison are shown. The guided condition produced two rejected candidates followed by one recorded acceptance. The
 444 declarations-only condition first terminated at the token limit with no visible proof body; this first control request is therefore
 445 truncated rather than a Lean rejection. The next two declarations-only candidates were rejected. Six requests and five
 446 candidates are one bounded comparison, not independent replications, and saved compilation acceptance is not upgraded here
 447 to new kernel certification.

448 **Methods**

449 **Study basis and evidence scope**

450 This study combines a conditional analytical examination of research continuation with retrospective
 451 analyses of retained coefficient-continuation and proof-completion records. The sources include dated
 452 theoretical and design documents, the original mathematical source, common task materials, generated
 453 candidates, compiler records and prior analyses. The analysis uses the preserved proof-completion
 454 archive, the separate coefficient-continuation archive and independently logged deterministic reanalysis.
 455 It adds no new model responses or experimental units. Newly executed checks are dated and logged
 456 separately from historical compilation and prior audits. Supplementary Notes 1, 5 and 6 distinguish each
 457 evidence identity.

458 **Theory and mathematical scope**

459 A future regime specifies duties, tool semantics, permissions, resource limits and the continuation policies
 460 being compared. Output equivalence is relative to a declared observation M , and future equivalence is
 461 relative to the chosen regime, including the outcome and cost measures. Neither is a claim about complete
 462 causal-state equality. For the finite witness, the observation is precisely r and the restricted policy is a
 463 deterministic real increment depending only on r . We do not assume that a language model obeys that
 464 restriction. The two targets differ because $p_{\pm} = z_{\pm}^2 + r$. The minimax result is an analytic derivation
 465 inherited from the earlier theoretical materials; its full proof and the state-aware sufficient construction
 466 are given in Supplementary Note 2. Symbolic checks and an illustrative deterministic figure are additional
 467 checks; the all-quantifier result is established by the piecewise proof.

468 The smooth cutoff construction has declared inner and outer rectangles. Positivity of the radicand and
 469 denominators establishes analyticity of the specified core functions on a neighbourhood of the outer
 470 rectangle. Transferring this local operation to the source wave classes additionally requires their fixed-
 471 stage, derivative, support and multiplier hypotheses. We do not claim that arbitrary generated expressions
 472 satisfy them, that physical norms improve monotonically, or that this rewrite independently proves a full

473 NS replacement theorem. Earlier source-relative PDE arguments, recorded computational checks and
 474 independently certified mathematics are separate evidence classes. Supplementary Notes 2–3 also state
 475 the sufficient-state boundary and its cost assumptions.

476 **Mathematical source and local target**

477 The source is the retained `NavierStokes/PrimaryCovarianceBounds.lean` at the recorded OpenAI repository
 478 commit. Its role is a local covariance bound within the published construction, not an original theorem of
 479 this study. The fixed outer theorem chooses uniform constants and a scale threshold before the pulse,
 480 chart parameter and target multiplier. Its local gap is the assertion that every entry of `normalizedPair P` is
 481 within ρ of $H_0 \rho$. The source identifier ρ is written ξ in the paper; `slotRadius r0 h n` is R_n ;
 482 `ChartScales.slotLength r0 h n` is L_n ; source coefficients E, D are written e, d ; and c is
 483 `PulseCovariance.concentrationConstant a A b B`. The source positivity, compactness, pulse estimates and
 484 column-mass results are premises of this task, not model outputs.

485 Both conditions received the exact outer template with one marked proof-body hole and the full relevant
 486 declarations of `normalizedPair_entry_error` and `slotRadius_sq`. The additional note in the guided condition
 487 identifies the needed interval/denominator/centre conversion, the integral-error application and the scalar
 488 comparison. It does not contain the final tactic body, but it does provide an explicit solution strategy. Full
 489 inputs, all five candidate bodies, the truncated-response status and the accepted downstream line are
 490 retained in Supplementary Note 5 and the evidence archive. No claim of equal full informational state is
 491 made.

492 **Selection, generation and feedback**

493 The final protocol was selected after researcher-directed modification of broader reconstruction
 494 workflows. The investigator explicitly authorised a known-proof comparison with the last local step left
 495 to the model. Both conditions used a maximum of three requests. The retained seed root was
 496 1,775,273,732, increased by attempt; the recorded arm order was guided then declarations-only. Equal
 497 attempt seeds do not imply identical stochastic trajectories under different prompts. Each later request

498 included the condition's own visible candidate and compiler feedback, when available. At most the final
 499 12,000 characters of each output stream were exposed as feedback, with truncation marked; complete
 500 streams were retained. Models returned Lean proof-body text without live shell, history-retrieval or
 501 search tools. The enclosing source was filled mechanically without semantic correction of the generated
 502 body.

503 The recorded serving alias was qwen3.8-27b-fp8-a256k-nomtp-c1, with local model-root basename Qwen3.8-
 504 27B-FP8. The deployment records specify 262,144-token context, one globally active generation, MTP
 505 disabled, half-precision computation, tensor parallelism 2, turboquant_k8v4 KV configuration, piecewise
 506 capture size 4 and safe synchronisation. Sampling used temperature 1, top-p 0.95, top-k 20, presence
 507 penalty 0, repetition penalty 1 and an active end-of-sequence rule, with recorded medium thinking. Per-
 508 request output was limited to 32,768 tokens including reasoning; input allowance was 196,608. These are
 509 historical configuration records, not independently verified current service properties or model weights.
 510 There was no imposed client generation deadline in this final run.

511 **Recorded acceptance and compilation**

512 Five non-empty proof bodies were inserted into the identical template and compiled in the retained Lean
 513 4.34.0-rc2 environment. The first declarations-only request had `finish_reason=length` and no visible body,
 514 so no compiler was invoked for that response. Recorded compilation took approximately 13–14 seconds
 515 per candidate. Non-fatal cache/permission warnings in the saved compiler outputs remain part of the
 516 evidence; the logs have not been cleaned into idealised feedback. The successful record contains a zero
 517 return code, the exact enclosing type check and the printed axiom dependencies. Failed outputs may list
 518 `sorryAx` because their proof remains incomplete; this is not evidence of deliberate use of an axiom to
 519 evade the task.

520 Record analysis reread the original handoff and reran the auditor in an isolated output directory. It
 521 verified the 859 payload identities, reconstructed all six response streams, matched the five candidate
 522 insertions and compiler receipts, and checked the 243 shared work-tree files outside the candidate

module. The original collector duplicated two mirrored reasoning fields; their bytes are retained and the duplication documented. Only visible proof bodies and compiler feedback entered later requests; server usage is the token source. These record checks were followed by offline recompilation of the five unchanged saved candidates under Lean 4.34.0-rc2 and the pinned dependencies. The ordinary Lean replay reproduced one acceptance and four rejections; the truncated request has no candidate to compile. This checks the saved candidates with the same compiler family, not with an independent kernel.¹⁹

Sources and supplementary checks

The Codex background refers to the supplied snapshot at commit a8964cb1bad67bc26a826fb07d1bef99c6a3f008, including the new-context handler and history-notes extension, backend and tools. The handler requests a new window without summarising the conversation. The extension exposes note and historical-content operations through a backend; the selected client files do not establish the server's internal storage or the behaviour of all releases. Public source semantics, our experimental implementation and opaque internal state are different evidence classes. The mathematical community's statement supplies a purpose-level argument, not measured general decline, and the NS process account is attributed to its publisher rather than treated as an isolated causal experiment. Two ancillary records have distinct provenance. The independent diagnostic is a separate Codex task with its own stated same-start materials and retained local compilation records; it was interrupted and did not complete the full NS goal. Its model and tool environment differ from the final comparison, and its input boundary is not independent certification of blindness. The researcher recheck is a reference-exposed derivation and algebra check; its own account explicitly records no new Lean compilation. Neither record is pooled with the formal six-request comparison.

Evidence interpretation and analysis

In the proof-completion study, the unit of observation is one selected obligation and its paired bounded attempt sequences. The six requests, five candidates, matrix entries and compiler messages are not independent samples. We report every attempt, the exact endpoint, visible tactic differences and realised

resources. No population success rate, p-value, confidence interval or general efficiency estimate is inferred. The guided prompt adds 332 initial input tokens and procedural information; feedback sequences diverge. Additional explanation, context arrangement, unequal realised effort and stochastic variation remain possible explanations of the observed contrast. Both final candidates locate the relevant lemmas, which supports a specific diagnosis of application failure but does not isolate a unique cognitive mechanism.

The passage from `hclose` to `hmargin` is an observed source-level use of a generated object by a supplied consumer. It does not show that the agent designed the downstream programme, changed its general method or adapted an acquired relation to a new duty. Earlier zero, mixed, stopped and unrun studies retain their original identities in Supplementary Note 6. The separately executed coefficient-continuation evidence is detailed in Supplementary Note 9. Neither its selected favourable traces nor this local proof completion replaces an original study endpoint. Supplementary Note 7 distinguishes observed reconstruction from still-unidentified causal and cross-domain claims.

561 Archived successor studies

The assistance study recorded four formation episodes with two shared archived ancestors, followed by two continuations in each of two feedback conditions. The adoption study recorded eight new formation episodes, a common assisted first continuation, and two subsequent version-adoption branches per episode. The totals are 20 and 32 stages. Later targets were fixed modifications of the preceding targets; each continuation read its actual predecessor's amplitude array and programme inventory. The agent wrote two real expressions inside a supplied smooth-cutoff update family. Both branches could retain or reconstruct accessible programmes. These were coefficient-workbench duties, not generation of full NS fields.

The completed-record analysis uses 40 assistance returns and 64 adoption returns. One additional assistance request remains historically unknown; a separately authorised supplementary observation occupies its analysis slot without recovering that request. No unit is added by this editorial reanalysis. The

complete stage counts include one empty final programme in assistance and four in adoption. The original protocol treated these as no applied update, even where a usable draft existed. This historical rule is preserved, not corrected after the fact. Initial and revised code were exactly equal in seven of eight adoption pairs. The original treatment-minus-control family contrasts and complete denominators are reported in Supplementary Note 9 rather than conditioned on valid final outputs.

Two traces, assistance episode P00 and adoption episode P07, are selected after examining the results for their inspectable mathematical content. The programmes, original visible text, feedback rows and parent-state references were reread directly from the archived files. A separate interpreter recomputed their displayed safe-core operations at 100-digit precision, while direct arithmetic checked all 18,720 terminal field rows. Forty inherited-state arrays were matched to predecessor outputs and source hashes. The few displayed coordinates are within-duty illustrations, not independent experimental units or estimates of average improvement. The whole-residual identities are derived analytically from the actual programmes. Their series properties do not license monotone improvement for every source, point or physical norm.

Data availability

Figure Source Data, the complete mathematical task inputs, all five saved proof candidates, six visible proof-completion responses and compilation records, and source-mapped coefficient-continuation records are included in the accompanying review-support archive, MA0FIELD_REVIEW_SUPPORT.zip. The successor material preserves all recorded terminal outcomes and original study denominators, with separately identified explanatory traces. Provenance maps identify every supplied excerpt and any privacy transformation. The full retained scientific raw records, with file-level provenance, are archived at <https://doi.org/10.5281/zenodo.22904604>. This broader sharing scope does not enlarge the analysis inputs, observation units or original denominators. The associated preprint is <https://doi.org/10.5281/zenodo.22905100>. Original data, text and figures are CC BY 4.0 and original code Apache-2.0, to the extent of the contributor's rights; third-party rights and notices remain in force.

597 Code availability

598 The review-support archive contains figure-generation sources, exact mathematical checks, raw-record
599 reanalysis, the successor-record audit and assembly/replay scripts for the five saved Lean candidates.
600 REPRODUCIBILITY/README.md identifies the input and output of each offline command and its verification
601 layer. Default entry points do not contact a model service. The one-command entry point uses packaged
602 Python and Lean 4.34.0-rc2 environments, verifies the raw-data identities, reruns the mathematical and
603 record checks, and recompiles the five unchanged saved candidates offline. Ordinary Lean replay has
604 been executed; independent-kernel verification remains unexecuted. Third-party notices accompany their
605 sources. The repository is <https://github.com/Wangziqu0/MaoField/tree/nature-main>, with the fixed
606 release tagged nature-v8-r3. Software, original study data and packaged reproduction environments are
607 archived at <https://doi.org/10.5281/zenodo.22905103>. Missing environment assets can be downloaded
608 from that fixed GitHub release and are hash-verified before use; mathematical analysis and candidate
609 compilation then run offline. The older main branch and R3.0 history remain distinct.

610 AI use and research attribution

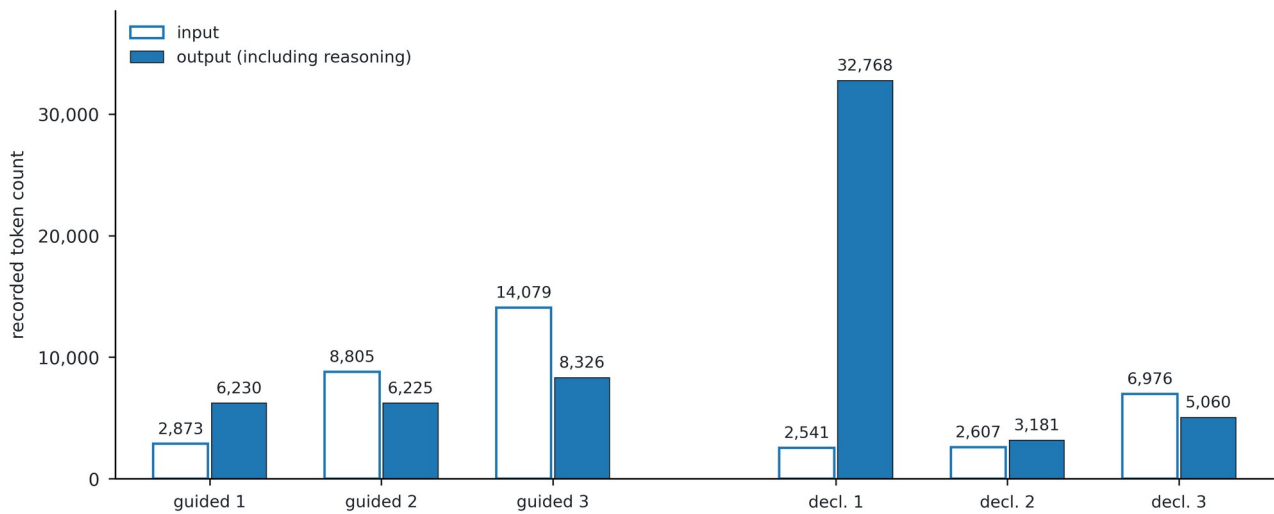
611 The archived studies used model-generated programmes or proof bodies, investigator-designed tasks and
612 guidance, and Codex-based execution. AI assistance was also used in source organisation, deterministic
613 analysis, drafting and figure preparation, with these outputs checked against the retained records and
614 mathematical arguments. This revision adds no new experimental model responses. Source-mathematics
615 authorship, investigator steering and checking, generated artefacts and computational verification are
616 distinct contributions. AI systems are not designated as authors.

617 Additional information

618 Supplementary Information is available for this paper.

619 **Extended Data**

Extended Data Fig. 1 | Recorded resources in the selected proof-completion comparison



The first declarations-only request was truncated and produced no visible proof body.
Counts describe realised usage only; they are not a general efficiency comparison.

620

621 **Extended Data Fig. 1 | Recorded token use, not efficiency.** The six input and output counts are reproduced from the supplied
622 source-data table. Output includes reasoning. The first declarations-only response reaches its configured limit without a visible
623 proof body. Guided inputs grow with their own candidates and compiler feedback. The figure describes realised work in this
624 one comparison; equal request limits did not equalise realised computation, and no general efficiency advantage is inferred.

Extended Data Table 1 | Earlier workflows retain their identities. Supplementary Note 6 lists the retained pre-hclose design history, including stopped or unrun branches. The coefficient-continuation and proof-completion rows are supported by raw-record reanalysis; older rows retain their historical-report identity. The completed successor studies are separately identified, not pooled with the proof-completion pair. Their failed, unchanged, truncated, unknown and NOT_RUN states do not become positive evidence because the final local completion was accepted.

Recorded workflow	Retained outcome	Evidence status
Full-endpoint workflow	Four branches; 55 requests; no writes	Earlier exploration
Adapted continuation	40 requests; no writes; PI stop	Stopped
Dynamic organisation	23 requests; no writes	Earlier exploration
Material preparation	16 preparation calls; packet not accepted; four branches unrun	Preparation failure
Fixed-material continuation	23 requests; no writes; three remaining branches unrun	Earlier exploration
Precompiled-source continuation	42 requests; no writes; stop	Stopped
Independent Codex diagnostic	Own specified same-start task; saved local compilation; PI stop	Separate diagnostic, not a formal effect control
Researcher recheck	Full reference seen; algebra recheck; no new Lean compilation	Reference-exposed verification
Final hclose pair	Six requests; five candidates; guided third accepted	Present local record
Coefficient assistance / adoption	52 stages; 104 completed returns. Actual revision and reuse; 5 empty finals; 7/8 equal adoption versions.	Executed; raw reanalysis. No established average benefit.